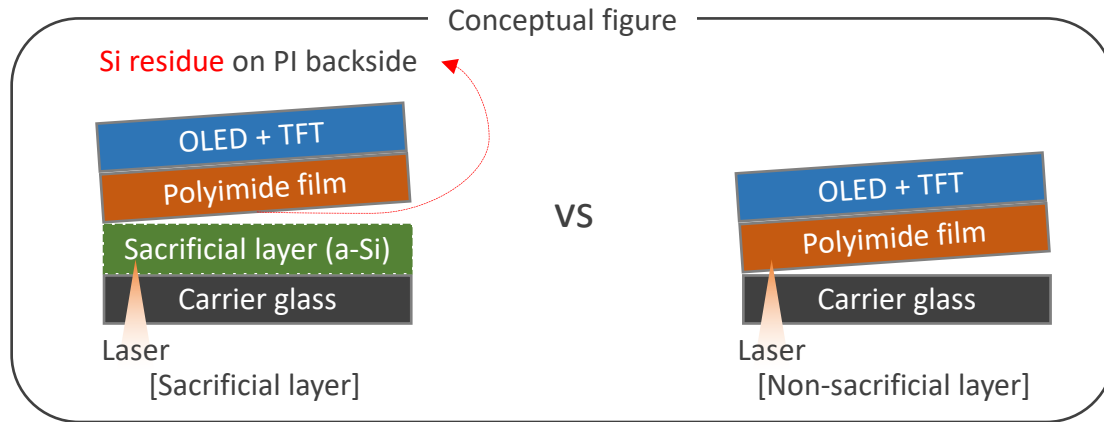


Validated Non-Sacrificial Layer Integration with Double-Digit Yield Improvement

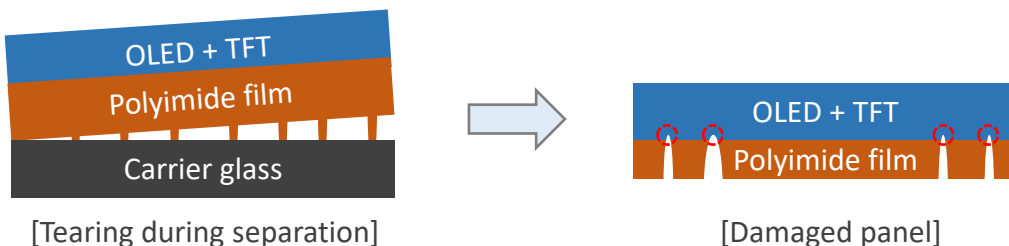
Objectives

- Non-sacrificial layer technology development
 - Improve under-display optical sensor performance



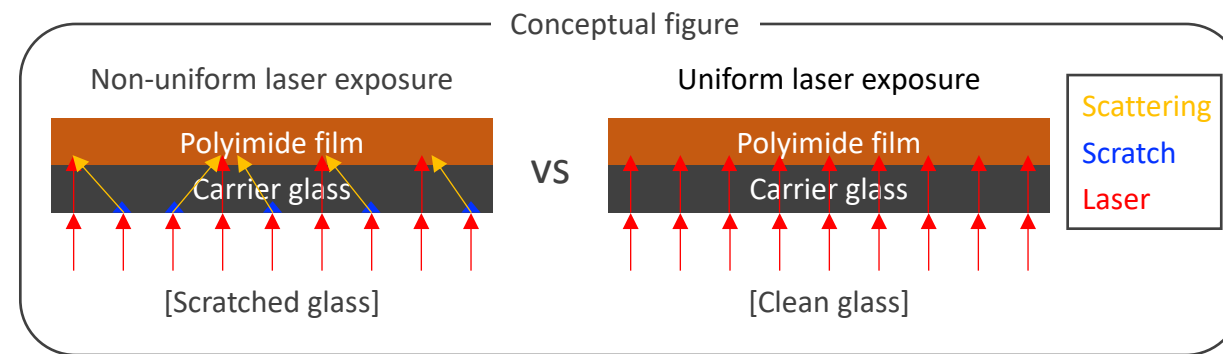
Challenges

- PI ash
 - Excessive localized laser energy damages PI, generating ash that degrades optical sensor performance
- Under-processing
 - Insufficient laser exposure leads to incomplete release, causing rough delamination and PI tearing



Approach

- Laser lift-off (LLO) process optimization
 - Optimize the laser source, beam profile, and key process parameters to ensure **uniform laser exposure**
- Laser-blocking scratch reduction
 - Reduce glass scratches to minimize **scratch-induced laser blocking**



- Implement backside inspection and statistical analysis to identify the **process steps driving glass scratches**
- Adjust key process parameters to mitigate **glass scratch formation**

Outcome

- Yield improvement
 - Achieved double-digit (%) yield improvement for non-sacrificial layer integration
- Backside quality improvement
 - Reduced PI backside ash and glass scratches